AI Cost Analysis — Summary

LegacyLens | Voyage Code 3, Claude Haiku/Sonnet, Supabase pgvector | March 2026

API Pricing

Service	Rate
Voyage Code 3 (embeddings)	$0.06 / 1M tokens
Claude Haiku 4.5 (fast mode)	$0.80 input / $4.00 output per 1M tokens
Claude Sonnet 4 (quality mode)	$3.00 input / $15.00 output per 1M tokens
Supabase Pro	$25/month

1. Development & Testing Costs

Category	Details	Cost
Embedding — ingestion	~4,000 chunks × 800 tok × 5 runs = 16M tokens	$0.96
Embedding — queries	Elbow analysis + eval (~320 queries)	<$0.01
LLM — Haiku (testing)	~250 queries, avg 4,247 in / 697 out tokens	$1.76
LLM — Sonnet (testing)	~150 queries, avg 6,537 in / 775 out tokens	$4.68
LLM — Related Qs	~100 calls (Haiku, 300 in / 100 out)	$0.06
Supabase hosting	Pro plan, ~2 months	$50.00
Vercel hosting	Hobby tier	$0.00
Total development cost		$57.66

Database hosting accounts for 86% of development spend. Total AI API cost was only $7.66. Embedding costs are negligible ($0.06/1M tokens). Re-ingesting the full codebase costs just $0.19 per run.

2. Production Cost Projections

Assumptions

5 queries/user/day × 22 weekdays = 110 queries/user/month. 70% fast mode (Haiku, 3 chunks), 30% quality mode (Sonnet, 5 chunks). 30% cache hit rate (5-min TTL). 1 monthly full re-ingestion.

Per-Query Cost

Component	Fast (Haiku)	Quality (Sonnet)
Query embedding	$0.000002	$0.000002
LLM input tokens	$0.0034	$0.0196
LLM output tokens	$0.0028	$0.0116
Related questions	$0.0006	$0.0006
Total per query	$0.0068	$0.0318

Blended cost (70/30 split): $0.0143/query. After 30% cache hits: $0.010/query (~1 cent).

Monthly Cost by Scale

	100 Users	1,000 Users	10,000 Users	100,000 Users
Queries/month	11,000	110,000	1,100,000	11,000,000
AI API cost	$132	$1,321	$13,212	$132,129
Infrastructure	$25	$45	$225	$800
Grand total	$157	$1,366	$13,437	$132,929
Per user/month	$1.57	$1.37	$1.34	$1.33
Per query	$0.014	$0.012	$0.012	$0.012

Infrastructure: Supabase Pro $25 (scales to Team $75 at 10K+, dedicated $300+ at 100K+). Vercel: free → Pro $20 → $150+ → $500+ at scale. At 10K+ users, migrate to distributed cache (Upstash Redis ~$10/mo) for better hit rates.

Cost Sensitivity — Fast/Quality Split

Fast / Quality Split	Per Query	1,000 Users/mo	10,000 Users/mo
100% / 0% (all fast)	$0.005	$527	$5,270
70% / 30% (baseline)	$0.010	$1,366	$13,437
50% / 50%	$0.014	$1,831	$18,005
0% / 100% (all quality)	$0.022	$2,896	$28,655

Key Optimization Levers

- Maximize fast mode usage (default): 4.7× cheaper per query than quality mode
- Increase cache TTL (5min → 15min): could raise hit rate to 50%, cutting API costs ~29%
- Distributed cache at scale: eliminates per-instance cold cache on Vercel serverless
- Re-ingestion is trivially cheap: $0.19/month for full re-embed, $0.02 for 10% incremental

Summary

Metric	Value
Total development cost	$57.66
AI API share of dev cost	$7.66 (13%)
Effective cost per query	~$0.01
Cost per user/month (1K users)	$1.37
Primary cost driver	LLM tokens (Sonnet quality mode)
Best optimization	Fast/quality split + cache TTL
Break-even SaaS price (1K users)	~$5/user/month (73% margin)